

under arbitrary scattering conditions. The main result is that the number of DoF per unit area is equal to the number of Nyquist samples per square meter to be able to reconstruct the field within a specified accuracy. The authors of [28] used the Fourier plane-wave series expansion for electromagnetic fields to evaluate the number of additional DoF in the reactive near-field, which is determined by the evanescent waves.

The author of [29] introduced a general approach to evaluate the number of effective communication modes and their strength between two volumes. The solution is cast in terms of an eigenproblem obtained from the wave equation and Green's function. An interesting result is that the number of DoF between two parallel communicating surfaces of area A_T and A_R , located at a distance r , is equal to $A_T A_R / (\lambda^2 r^2)$, provided that the two surfaces are in the paraxial regime. Under the paraxial setting, to elaborate, r is such that the Fresnel approximation can be applied, that is, only the first two terms of the Taylor series expansion of r are retained, and the surface apertures are small compared to the propagation distance. Following a similar methodology, the authors of [30] derived a closed form expression for the DoF of LoS channels as a function of the orientations of the arrays, while still assuming the paraxial setting.

Naturally, analysis of the DoF requires, especially in free-space channels, careful consideration of the transmission distance, the size of the surfaces, and the relative geometry of the surfaces. Thus, when the paraxial approximation, which is often assumed in the literature, is not fulfilled, that is, the transmission distance is comparable with the sizes of the surfaces, the calculation of the DoF is not straightforward, and alternative methods are required. More specifically, besides the numerical solution of the eigenproblem formulated in, e.g., [29], the DoF can be obtained by applying two general methods, which are known as (1) the spatial bandwidth [31]-[34] and (2) Landau's methods [35]-[37], with the latter being highly coupled with the eigenproblem in [29]. Even though these methods are quite general and can be applied to general network scenarios, including non-paraxial settings, explicit analytical expressions for the DoF are difficult to obtain in general network deployments, e.g., in non-paraxial settings. Approximation approaches based on the spatial bandwidth and Landau's eigenproblem, which can be applied to non-paraxial settings, can be found in [12] and [37], respectively. The impact of spatial blocking on the DoF has recently been analyzed in [38]. A short overview based on [4] and applied to continuous linear arrays can be found in [39]. To overcome these limitations, some ad hoc methods have recently been proposed.

For example, the authors of [40], have provided closed-form solutions for the DoF between a linear LIS and a linear small intelligent surface (SIS), which is of practical interest since the considered setting models the link between a base station and a mobile terminal. While the approach in [40] is a key step towards understanding non-paraxial settings, a limited geometric setup is considered, and the solutions are restricted to specific conditions.

In the present paper, we are mainly interested in linear continuous arrays. In this context, the authors of [41] have

recently introduced an analytical approach to compute the DoF based on the spatial bandwidth method. A crucial assumption in [41] is that the considered linear arrays are composed of an infinite number of ideal point sources and that the transmitting linear array is characterized by an azimuth-independent radiation pattern. Under these assumptions, it is sufficient to parameterize the relative direction of the receiving linear array by using a single polar angle. This allows for a simplified analysis, since the transmitting and receiving linear arrays are always in the visibility region of one another. The same authors generalize the analysis in [42], by introducing analytical asymptotic expressions for spatial bandwidth, highlighting the importance of the network geometry on the DoF.

Although DoF evaluation has been investigated in conventional geometric settings, especially in paraxial setup, a comprehensive performance analysis remains elusive, especially from a system-level (or statistical) point of view [43]. Using fundamental results from stochastic geometry theory, new spatial models and channel statistics can be conceptualized to capture the peculiarities of near-field communications. This will allow for the characterization of location-dependent DoF in the receiver's association policy, with major applications in network performance analysis.

In the present work, a new analytical framework for calculating the DoF of a communication link between a large continuous linear array, and a small intelligent line, still modeled as a continuous linear array, is introduced. The proposed approach can be applied to paraxial and non-paraxial settings. To this end, the arrays are modeled as electromagnetic lines in free space, obeying the Huygens-Fresnel principle. The two arrays are assumed to radiate only in one of the two half-planes, in order to consider realistic deployments in which the transmitting array radiates only towards the intended receiving array. The transmitting array uses a steering function to concentrate the energy towards the direction of a focal point on the receiving array in the far-field. The calculation of the DoF is based on the calculation of the number of orthogonal steering functions that fit within the length of the receiving array. This approach allows us to account for near-field channels and non-paraxial settings. Therefore, the effective lengths of the two mutually visible arrays play an important role in the calculation of the DoF, since the mutual visibility of the arrays also determines the steering functionality. Another factor that determines the DoF is the projected lengths of the two linear arrays onto the line that is perpendicular to the line that connects the centers of the two arrays. These projections are highly dependent on the positions and orientations of the arrays.

To elaborate, the contributions of this paper are summarized as follows:

- For a pair of linear arrays lying on the same plane, we provide a methodology to characterize the complete or partial visibility conditions between the arrays, without any restrictions on their relative positions and orientations. The considered unconstrained geometric setting, in which the lines can transmit only in a half hemisphere, can, in fact, lead to the lack of visibility or to a partial visibility condition. In the latter case, the effective length